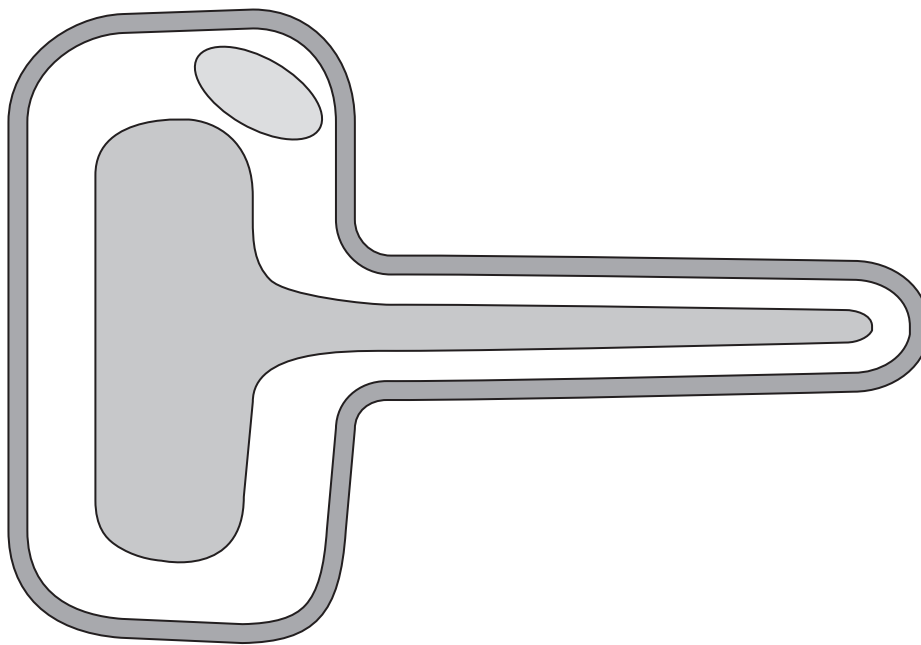


Answer ALL questions.

Write your answers in the spaces provided.

Some questions must be answered with a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

- 1 The diagram shows an incomplete plant cell, as drawn by a student.



- (a) (i) Label the **tonoplast** on the diagram.

(1)

- (ii) Draw **and** label a **nucleolus** in this plant cell.

(1)

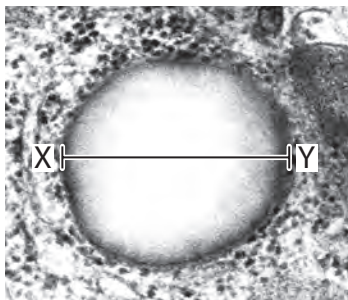
- (b) Complete the table to show which of these structures would be present **or** absent in plant leaf cells.

(2)

Structure	✓ if present × if absent
amyloplast	
chloroplast	
Golgi apparatus	
plasmodesmata	



(c) The photograph shows a lysosome, as seen using an electron microscope.



(Source: © Science History Images / Alamy Stock Photo)

Magnification $\times 250\,000$

Calculate the actual width (X–Y) of this lysosome in μm .

Give your answer to **two** decimal places.

(1)

Answer μm

(Total for Question 1 = 5 marks)

